

AI Persona Web Application using Gemini API

1. Objective

Develop a **web-based AI Persona Application** using the **Gemini API**.

The application should allow users to interact with different AI personas and receive responses according to the selected persona's role, audience, context, format, constraints, and language.

The main focus of this deliverable is:

Frontend Development + Gemini API + Persona Prompt Engineering + Multiple Persona Responses

2. Important Instructions

- Minimum **4 personas** are required.
 - The user must be able to select **one or multiple personas**.
 - Responses must be generated using the **Gemini API**.
 - Different personas should provide **meaningfully different perspectives**.
 - For multiple personas, **prefer one Gemini API request containing all selected personas** rather than making unnecessary separate requests.
 - Use the six Prompt Card elements:
 - Role
 - Audience
 - Context
 - Format
 - Constraints
 - Language
 - The application must be contained in **one HTML file** containing HTML, CSS and JavaScript.
 - Do not publish your actual Gemini API key on GitHub.
 - The application must handle basic input/API errors.
 - The submitted application must be functional and demonstrated in the video.
 - Your GitHub repository must contain a clear README.
 - Deliverables are: Demo video link of your project and Github link with no api key
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3. Application Theme

Your application will be designed as an: AI Career Counsellor

The application should provide career guidance from different AI career counsellor personas. You must create **at least 4 different personas**.

Suggested Personas: You may use these personas or create your own:

1. **Technical Career Counsellor**

- AI/ML
- Programming
- Software development
- Technical skills
- Projects

2. **HR & Placement Counsellor**

- Resume
- Interviews
- Employability
- Recruitment
- Placement preparation

3. **Academic & Research Counsellor**

- Higher studies
- MS/M.Tech
- PhD
- Research
- Certifications

4. **Entrepreneurship Counsellor**

- Startups
- Business ideas
- Freelancing
- Product development
- Entrepreneurship

You may create additional or completely different personas, provided that each persona has a **clearly different perspective**.

4. User Interaction

The application must allow the user to:

A. Ask One Persona: The user selects one persona and enters a question.

Example:

"I know Python and basic Machine Learning. What should I learn to become an AI Engineer?"

The selected persona should provide the response according to its defined persona. Questions other than the persona should say I don't know.

B. Ask Multiple Personas: The user must also be able to select multiple personas and ask the same question.

Example:

None

Select Persona(s):

- ☐ Technical Career Counsellor
- ☐ HR & Placement Counsellor
- ☐ Academic Counsellor
- ☐ Entrepreneurship Counsellor

The user may select:

None

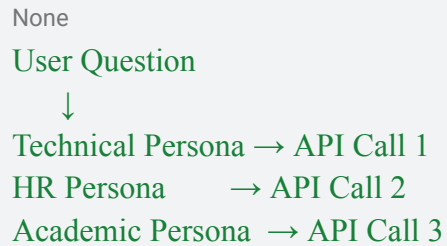
- ☒ Technical Career Counsellor
- ☒ HR & Placement Counsellor
- ☒ Academic Counsellor

The application should then generate a separate response from each selected persona.

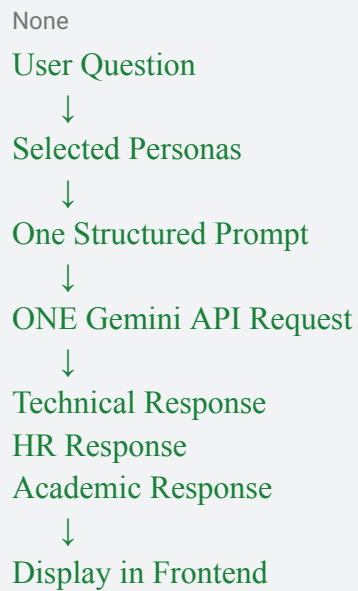
5. Important API Requirement

When multiple personas are selected, you should **preferably make one Gemini API request containing all selected personas**, rather than making a separate API request for every persona.

Avoid:



Preferred:



This is important because API services have **request/rate limits**. Your implementation should avoid unnecessary API calls.

6. Persona Prompt Card

Every persona must be designed using the following **six Prompt Card elements**: These six components should be used to construct your persona prompt.

Prompt Card Element	Meaning
Role	Who should the AI act as? What persona should it adopt?
Audience	Who is the AI giving the response to?
Context	What background information should the AI consider?
Format	How should the AI present its response?
Constraints	What rules or limitations should the AI follow?
Language	Which language should the AI use?

7. Prompt Card Example

For example, we are celebrating **National Space Day** and have already prepared a poster. We want to upload it on our Instagram account. To generate an appropriate caption, we can create a Prompt Card using the six elements.

Sample Prompt Card

Prompt Card Element	Example
Role	Social Media Manager
Audience	University/college students
Context	We are celebrating National Space Day and have already designed an Instagram poster for the occasion. Write a

	caption to accompany the poster and celebrate India's achievements in space with ISRO.
Format	Instagram caption + hashtags
Constraints	Under 25 words, simple and engaging, no emojis, no jargon
Language	Simple English, no hard vocabulary

The complete Prompt Card can therefore be written as:

Role: Social Media Manager
Audience: University/college students
Context: We are celebrating National Space Day and have already designed an Instagram poster for the occasion. Write a caption to accompany the poster and celebrate India's achievements in space with ISRO.
Format: Instagram caption + hashtags
Constraints: Under 25 words, simple and engaging, no emojis, no jargon
Language: English

Important

This example is only to explain **how to create a Prompt Card**.

For your project, you must create Prompt Cards for your **AI Career Counsellor personas**.

8. Example Career Counsellor Prompt Card

Persona: Technical Career Counsellor

Prompt Card Element	Example
Role	Senior Technical Career Counsellor specializing in AI, ML and Software Engineering
Audience	Undergraduate ICT/Computer Science students

Context	The student is interested in AI/ML and wants guidance about skills, projects and career opportunities
Format	Recommendation + Skills to Develop + Project Suggestions + Career Roadmap
Constraints	Give practical and realistic advice. Do not guarantee jobs or salaries. Avoid unsupported assumptions.
Language	Simple English

Students should create similar Prompt Cards for their other personas.

9. Persona Differentiation

The personas should provide **different perspectives**.

For example, consider the question:

"Should I prepare for placements or pursue higher studies?"

Technical Career Counsellor

May focus on:

- Programming skills
- DSA
- AI/ML
- Projects
- Technical preparation

HR & Placement Counsellor

May focus on:

- Resume
- Interviews
- Communication
- Internships
- Employability

Academic Counsellor

May focus on:

- Research
- MS/M.Tech
- PhD
- Academic interests
- Higher-study preparation

Entrepreneurship Counsellor

May focus on:

- Startup ideas
- Product development
- Business opportunities
- Market validation
- Entrepreneurship skills

Simply changing the persona name while generating similar responses will **not demonstrate effective persona engineering**.

10. Prompt Construction

Your application should combine the Prompt Card information with the user's question.

The basic flow should be:

None

Role

+

Audience

+

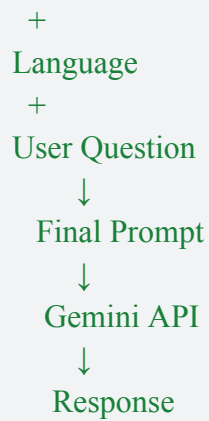
Context

+

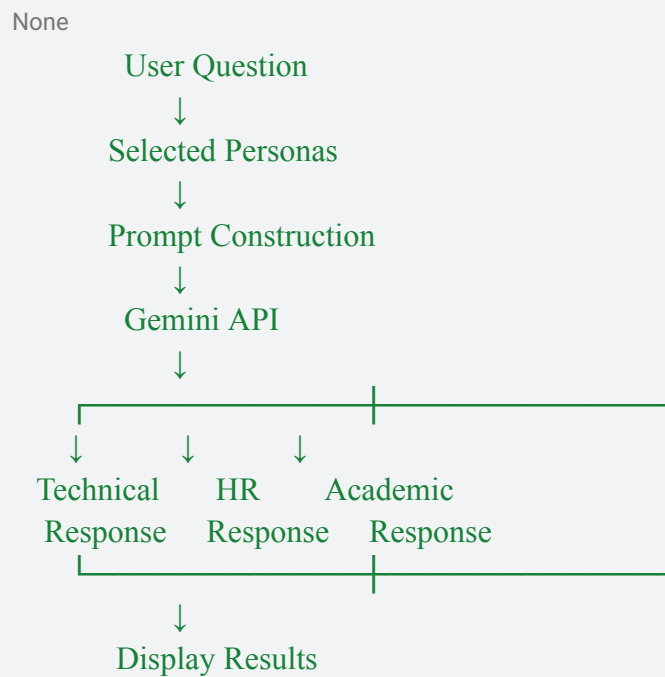
Format

+

Constraints



For multiple personas:



11. Required Frontend

Your application must contain:

1. User Question
2. A text area/input field where the user enters the question.

3. Persona Selection
4. The user should be able to select one or multiple personas.
5. Generate Button
6. A button such as: **Get Career Advice**
7. Response Area
8. Display the generated response clearly.
9. For multiple personas, display each response separately.

Example:

None

Technical Career Counsellor

[Gemini-generated response]

HR & Placement Counsellor

[Gemini-generated response]

Academic Counsellor

[Gemini-generated response]

12. Comparison

When multiple personas are selected, the application should make it easy for the user to compare their recommendations. At minimum, display the responses separately. A comparison section is encouraged.

For example:

Aspect	Technical	HR	Academic
Main Recommendation	AI Job	Placement	Higher Studies
Priority	Technical Skills	Employability	Research
Suggested Action	Build Projects	Prepare Interviews	Explore MS/M.Tech

The comparison may be generated using Gemini or implemented using your own frontend logic.

13. Technical Implementation Requirement

The complete application must be created using **ONE HTML FILE**. Your submission must contain:

None
`index.html`

The HTML file must contain:

- HTML
- CSS
- JavaScript

inside the same file.

Required structure

```
HTML
<!DOCTYPE html>
<html>

<head>

  <title>AI Career Counsellor</title>
```

```
<style>
  /* CSS */
</style>

</head>

<body>

  <!-- HTML / User Interface -->

  <script>
    // JavaScript
  </script>

</body>

</html>
```

You should **not** create separate files such as:

```
None
index.html
style.css
script.js
```

All three must be included in the **single HTML file**.

14. JavaScript Responsibilities

The JavaScript inside your HTML file should handle:

- Persona selection
- User question
- Prompt construction
- Combining Prompt Card elements
- Gemini API request

- Receiving the response
 - Displaying responses
 - Handling multiple personas
 - Basic error handling
-

15. Gemini API

The actual AI-generated responses must come from the **Gemini API**.

Do not use hard-coded responses such as:

```
JavaScript
if (persona === "Technical") {
  response = "Learn Python and Machine Learning.";
}
```

The response must be generated by Gemini based on the prompt.

16. API Key Security

Never submit or publish your actual Gemini API key in a public GitHub repository.

Do not expose your API key in the submitted source code.

Do not commit:

```
None
.env
```

or any file containing your actual API key.

If a temporary key is used for demonstration, remove it before uploading the project to a public repository.

17. Error Handling

Your application should handle basic situations such as:

No persona selected

Display:

Please select at least one persona.

Empty question

Display:

Please enter your career-related question.

API error

Display an appropriate message instead of allowing the application to crash.

18. Testing

Test your application using at least **3 different questions**.

For example:

Question 1

Should I prepare for placements or pursue higher studies?

Question 2

I know Python but do not have any projects. What should I do?

Question 3

Should I become an AI Engineer, Data Scientist or Software Developer?

For each question, test:

- One persona
- Multiple personas

- Different combinations of personas
-

19. Project Demo Video

Create a video demonstrating the working application.

The video should show:

1. Application interface.
2. Available personas.
3. Selection of one persona.
4. User question.
5. Generated response.
6. Selection of multiple personas.
7. Same user question.
8. Multiple persona responses.
9. Comparison, if implemented.
10. Brief explanation of the Prompt Card and prompt flow.

The video should demonstrate the **actual working application**.

20. GitHub Repository

Your GitHub repository should contain at least:

```
None
project/
├── index.html
├── README.md
└── assets/
```

The `index.html` file must contain the complete HTML, CSS and JavaScript implementation.

README should include:

1. Project title

2. Problem statement
 3. Objective
 4. Personas
 5. Prompt Cards
 6. Technology used
 7. Gemini API integration
 8. Application screenshots
 9. How to run the application
 10. Sample questions
 11. Sample outputs
 12. Team members
-

21. Final Submission

Submit the following:

1. GitHub Repository: Complete project source code.
2. AI Persona Web Application Demo: Video demonstrating the working application.
3. Prompt Card: One-page Prompt Card showing the six components:

Role + Audience + Context + Format + Constraints + Language

22. Core Learning Outcome

The purpose of this assignment is **not simply to create a chatbot using Gemini**.

You should demonstrate that you can:

Design a Persona → Define the Audience → Provide Context → Control the Output Format → Apply Constraints → Specify Language → Add User Input → Generate a Persona-Specific Response using Gemini

For multiple personas:

**One User Question → Multiple Persona Instructions → Efficient Gemini API
Request → Multiple Perspectives → Meaningful Comparison**

The quality of your project will be judged primarily by **how effectively your prompt design**